

Qontint WebApp - Implementation & Feature Report

Detailed Breakdown of Implemented Features and Underlying Tech Stack

This report outlines the actual, user-facing features currently working in the Qontint web application and details the specific technologies and infrastructure powering each of these features.

1. Content Analyzer (Analyze Page)

Feature: AI-Driven NLP Content Scoring

What it does: Takes user content, a target keyword, and a selected industry vertical to analyze it for novelty, authority, and predicted search engine ranking.

How it runs: Frontend: React with Framer Motion for smooth transitions.
Backend: FastAPI endpoint (/api/v1/analyze) orchestrates several services:

- spaCy (M2 Module) extracts entities and maps Subject-Verb-Object triples.
- Neo4j Graph Database (M3 Module) compares entities to establish Authority.
- DuckDuckGo + httpx + Celery (M1 Module) scrapes SERP baselines.
- GradientBoostingRegressor (M6 Module) predicts SEO positioning.

2. Entity Authority Graph Explorer (Graph Page)

Feature: 3D Taxonomy Knowledge Graph

What it does: Visualizes semantic relationships and entity taxonomy in an interactive 3D graph. Users can filter by vertical (e.g., SAP Supply Chain), hover for authority scores, and see connections.

How it runs: Frontend: Uses Three.js and @react-three/drei for GPU-accelerated 3D ambient visuals and particle systems. Styled with premium glassmorphism.
Backend: Neo4j graph database queried over FastAPI endpoints (/api/v1/graph/snapshot). Calculates PageRank algorithms to supply Authority weights.

3. Dynamic Content Generator (Generate Page)

Feature: SEO-Optimized Content Generation

What it does: Allows users to generate new SEO-optimized content based on specific keywords to quickly fill content gaps identified in the analyzer.

How it runs: Frontend: Interactive React form with real-time streaming feedback.
Backend: FastAPI (/api/v1/generate) integrates with either local Ollama (running llama3.1:8b) or Gemini APIs. The pipeline is optimized to eliminate redundant database operations and maintain a 10-15 second response target.

4. Keyword Taxonomy Management (Keywords Page)

Feature: Vertical-Specific Keyword Browsing

What it does: Allows users to browse and manage the 116-keyword taxonomy across different B2B verticals like Accounting & Finance and Banking & Lending Ops.

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How it runs: Frontend: React list components with TailwindCSS filtering and dynamic states.
Backend: Fetches structured keyword and taxonomy data from a PostgreSQL relational database via FastAPI endpoints (/api/v1/keywords).

5. Query Intelligence Explorer (Query Intel Page)

Feature: B2B Fintech Query Intelligence System

What it does: A system to discover high-intent query sweet spots and explore SERP features for specific vertical searches.

How it runs: Frontend: Animated React UI showing data tables and insights.
Backend: Powered by the M1 SERP Collector running as background tasks via Celery and Redis, storing historical intelligence data in PostgreSQL for rapid retrieval.

6. Analytics Dashboard (Dashboard Page)

Feature: Overview Metrics & Visualizations

What it does: Provides an overview of all SEO and content metrics processed by the system through visual charts and summary cards.

How it runs: Frontend: Built with React and Recharts library for interactive SVG charting. Uses Framer Motion and GSAP for micro-animations.
Backend: Aggregation endpoints in FastAPI pulling summarized statistics from PostgreSQL.